Suppose we are given a pair of functions f and F analytic in regions Ω and Ω', respectively, with $\Omega \subset \Omega'$. If the two functions agree on the smaller set Ω, we say that F is an **analytic continuation** of f into the region Ω'. The corollary then guarantees that there can be only one such analytic continuation, since F is uniquely determined by f.

5 Further applications

We gather in this section various consequences of the results proved so far.

5.1 Morera's theorem

A direct application of what was proved here is a converse of Cauchy's theorem.

Theorem 5.1 *Suppose f is a continuous function in the open disc D such that for any triangle T contained in D*

$$\int_T f(z)\,dz = 0,$$

then f is holomorphic.

Proof. By the proof of Theorem 2.1 the function f has a primitive F in D that satisfies $F' = f$. By the regularity theorem, we know that F is indefinitely (and hence twice) complex differentiable, and therefore f is holomorphic.

5.2 Sequences of holomorphic functions

Theorem 5.2 *If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.*

Proof. Let D be any disc whose closure is contained in Ω and T any triangle in that disc. Then, since each f_n is holomorphic, Goursat's theorem implies

$$\int_T f_n(z)\,dz = 0 \qquad \text{for all } n.$$

By assumption $f_n \to f$ uniformly in the closure of D, so f is continuous and

$$\int_T f_n(z)\,dz \to \int_T f(z)\,dz.$$